

ELEC5514 Lab6: LoRa

Communications

This week's experiment will be divided into three sections:

1. Use the LoRa board to establish a connection with the gateway and send a message to the LoRa gateway.
2. Use the keyboard to enter messages and send them to the LoRa gateway.
3. Messages are continuously sent to the gateway to confirm the maximum distance of message transmission within the campus environment.

Section 1

Open the project:

SourceCode/Lab6_Lora/Projects/Multi/Applications/LoRa/End_Node/MDK-ARM/STM321073RZ-Nucleo/lora

In this section, you are asked to implement a simple communication between the Lora board and the gateway. You need to send your group number as a message to the gateway, like ‘*Group1*’. In this time, you can create this message directly in your code.

The following figures is an example by sending ‘*Hello*’, and you also can use another way to achieve this function:

```

VERSION: 44011000
txDone
rxTimeOut
rxTimeOut
txDone
rxDone
JOINED
[Tx] HELLO
txDone
rxDone
[LoraRxData] HELLO
LED OFF
[Tx] HELLO

```

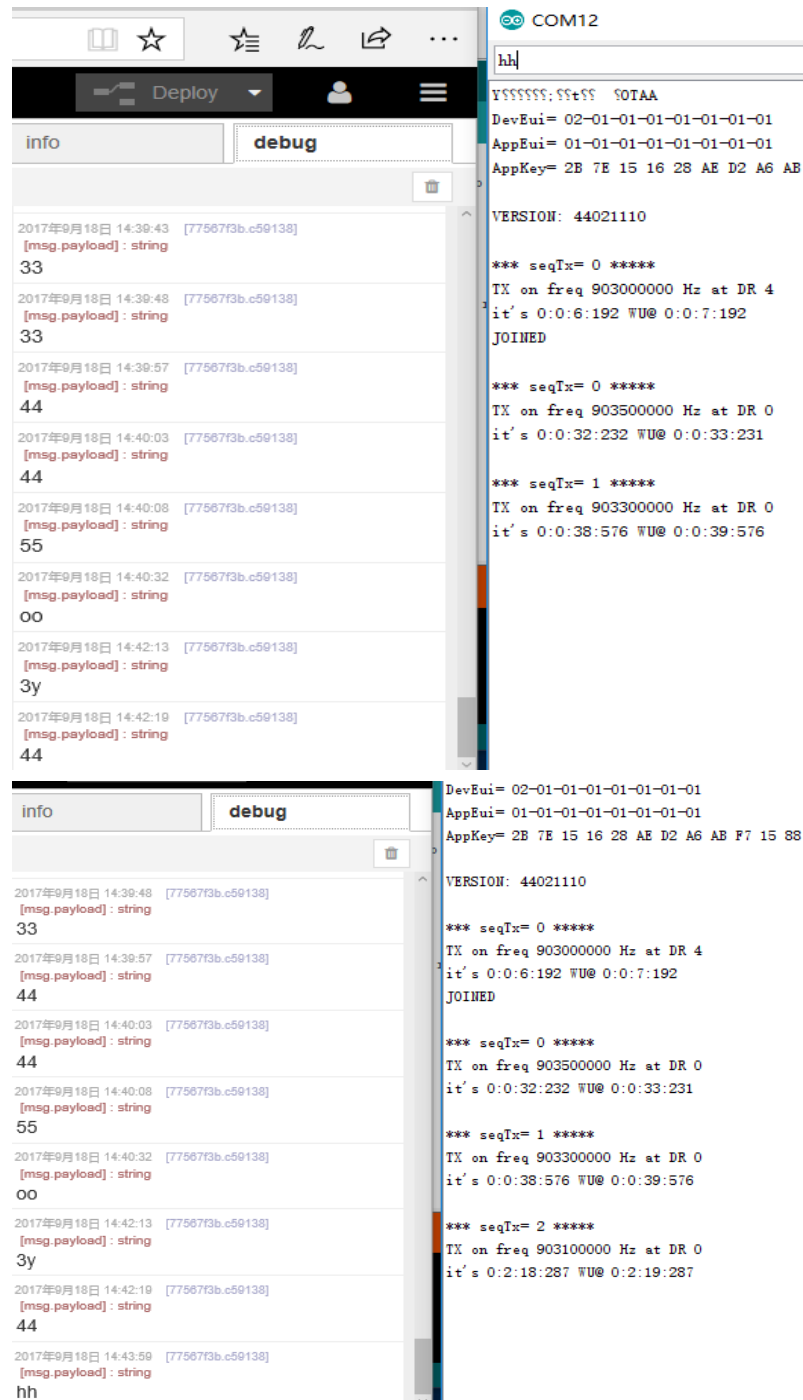
Tips: You can check the functions of *LoraTxData* and *LoraRxData* to understand the transmission and receive process. Note that the key word “*AppData*” has already been defined in this file and it is the set of the transmission information. You can read the codes from line around 219 for help, such as “*AppData->Port*”, “*AppData->Buff*” and “*AppData->BuffSize*”.

When you finish Section 1, call your tutor to check you result. You need to take the screenshot of the code you edited and your results for use in your report.

Section 2

In this section, you are required to input message (a character or a string) through the keyboard and send it to the gateway. In this section, you do not need to care about the Rx procedure.

The graph below shows an example. The left window is the gateway panel, which can display the message the gateway received. From the graphs, you can see at the gateway side, it can receive the typed in message “hh”:



Hints:

- 1) Enable the **UART initialization**. (Please pay attention to this case. It is not an easy task. If you have no idea how to achieve it, you can review the *BLE – Lab 4 code*)
- 2) Set a variable to store the typed message.
- 3) Call the “*LoraTxData()*” function to transmit this message.

Once you finished, please come forward with your programmed device. We will check the result for you. You need to take the screenshot of the code you edited and your results for use in your report.

Section 3

In this section, the students are required to measure the longest transmission distance for LoRa device. To achieve this goal, **you may need to modify your code.**

To do this, each group should prepare a laptop, which can provide the power and input message of the LoRa device, or you find your own way to let the device send the message. When the team arrives the longest distance point, you can calculate the relative distance from the point to our lab classroom, where the gateway is, and send the actual value distance to the gateway, and then there is one group number should stay at the classroom and record the number.

As we know, the LoRa enables more than 10km transmission, but maybe in the real transmission environment, it cannot provide that long range, so in the lab report, you can record your group longest value and analysis the reason for the distance.

When you finish Section 3, call your tutor to check you result. You need to take the screenshot of the code you edited and your results for use in your report.